Leaderboard

ROGII - Wellbore Geology Prediction

Submit Prediction

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

1564	Rafael Cunha	11.362	2	6d
1565	Mario Nicolae	11.363	3	1mo
1566	HongweiLuan	11.381	4	6d
1567	cage	11.411	4	1mo
1568	TAKASHI AZUMA	11.415	15	14h
1569	noahrolandd	11.422	8	1mo
1570	Orange Panda	11.455	2	44m

Your Best Entry!
This is your best entry, but your most recent submission is still being scored...

1571	Aristeides	11.465	30	8d
1572	不要玩手机	11.472	9	5h
1573	Promissa	11.475	5	4d
1574	Vcp1919	11.504	15	17d
1575	KK172	11.516	8	5d
1576	Khalid Mustafin	11.525	5	23d
1577	Kaloyan Stoyanov	11.544	1	7d

Report

Much of our work was modelled off of the procedures from the current public top scored notebook.

We built a machine learning pipeline to predict True Vertical Thickness (TVT) along the hidden evaluation zone of horizontal wells. Our methodology starts by treating each well as a separate sequence problem instead of a standard flat table problem. For every well, the pipeline reads the horizontal well file and its matching typewell file. The horizontal file provides measured depth (`MD`), spatial coordinates (`X`, `Y`, `Z`), gamma ray (`GR`), and `TVT_input`, while the typewell file provides known TVT, gamma ray, and geological labels. The `TVT_input` column is used to split the well into a known prefix and an evaluation zone. The known prefix is where interpreted TVT values are available, and the evaluation zone is where the model must predict TVT. The pipeline builds training rows mainly for the evaluation-zone rows, which matches the hidden test structure and reduces unnecessary computation. The known prefix is then used as an anchor by extracting the last known TVT, measured depth, and spatial position before prediction start. From this anchor, the code creates trajectory features such as distance from the last known point, changes in `X`, `Y`, and `Z`, horizontal displacement, and measured-depth distance into the evaluation zone. It also estimates TVT trends from the known prefix using slope-based baselines over the full known interval and over recent tail windows. These features estimate how TVT would continue if the known geological trend persisted.

Gamma ray data is handled as a major geological signal because changes in GR often correspond to changes in lithology and geological position. Missing GR values are interpolated when possible, and the pipeline generates rolling means, rolling standard deviations, first-order differences, second-order differences, and lag/lead features. These features describe both the local value and the shape of the GR curve. The pipeline also performs gamma ray signature matching using normalized cross-correlation. It compares unknown GR behavior after prediction start against the known GR and TVT pattern before prediction start across multiple window sizes. Conceptually, this asks which part of the known GR signature most closely resembles the current unknown GR window. The matched TVT estimates and correlation scores are then used as model features. The notebook also handles training-only geological formation columns such as `ANCC`, `ASTNU`, `ASTNL`, `EGFDU`, `EGFDL`, and `BUDA`. These raw columns exist in training data but not in hidden test data, so direct use would create leakage and train-test mismatch. Instead, the pipeline builds spatial formation imputers from the training wells. These imputers learn approximate formation surfaces from training-well `X` and `Y` locations and formation depths, then estimate comparable surfaces for both train and test wells. During training, the current well is excluded from its own imputation to reduce self-leakage. The imputed surfaces are then calibrated using each well's known TVT prefix by estimating a well-specific offset between known TVT, `Z`, and the imputed formation surface. This creates shared train-test features such as imputed surface depths, surface-minus-depth values, calibrated formation-based TVT estimates, and formation spread statistics.

The target is designed as a relative displacement rather than an absolute TVT value. Instead of training the model to predict raw `TVT`, the pipeline predicts `TVT - last_known_tvt`. This reframes the problem as predicting how far the geological position moves from the last known interpreted point. This is useful because absolute TVT values can vary across wells, while post-start movement from the known anchor is more comparable. At inference time, the predicted delta is added back to `last_known_tvt` to recover the final absolute TVT prediction. The model uses LightGBM regression with GroupKFold validation grouped by well. Grouping by well prevents rows from the same well from appearing in both training and validation sets, which avoids overly optimistic validation scores caused by row-level leakage. Before training, the feature list is built from the intersection of columns that exist in both processed training and test data, excluding identifiers such as `id`, `well`, and the target. This ensures every training feature can also be generated at prediction time. The feature matrix is cleaned by replacing infinite values, filling missing values, and casting numeric columns to `float32` for memory and runtime efficiency. During prediction, each fold model outputs a TVT delta for the test rows, the fold predictions are averaged, and the averaged delta is added to `last_known_tvt`. The final predictions are merged with `sample_submission.csv` using the required `id` field and saved as `submission.csv` with the required `id,tvt` format. It seems that most of the top submissions use a combination of models, publically available models, and extra data to produce a significantly lower RMSE result. Given the interest in time and resources and the amount of domain knowledge needed to achieve similar results, we are happy with our current attempt.